

Demand Estimation 4

PhD Industrial Organization

Nicholas Vreugdenhil

Plan

1. Estimation algorithm
2. Instrumental variables
3. Extensions
4. Applications

Plan

1. **Estimation algorithm**
2. Instrumental variables
3. Extensions
4. Applications

Review: setup of the problem

- (Conditional, indirect) utility:

$$u_{ijt} = x_{jt}\beta_{it} + \alpha_{it}p_{jt} + \zeta_{jt} + \epsilon_{ijt}$$

- (Note: first element of x_{jt} is 1 $\rightarrow$ absorbs mean of ζ_{jt})
- What are the parameters we need to estimate?
- Linear parameters:
 - Parameters from the mean utility equation: (α_0, β_0)
- Nonlinear parameters
 - Γ : coefficients on (observed) demographics
 - Σ : idiosyncratic “taste for characteristics”
- So, full parameter vector to estimate: $\theta = (\alpha_0, \beta_0, \Gamma, \Sigma)$.

Review: estimation algorithm: overview

- Step 1: For a guess of Γ and Σ , and a vector of mean utilities δ_t , compute model-predicted market shares.
- Step 2: For a guess of Γ and Σ do an **inversion**: find δ_t where the model-predicted market shares match the empirical market shares s_t .
 - This step will repeatedly call the function from Step 1.
- Step 3: Use the computed δ_t from Step 2 to compute $\xi_{jt} = \delta_{jt}(\Gamma, \Sigma) - x_{jt}\beta_0 - \alpha_0 p_{jt}$.
 - Interact with IVs to get the GMM objective function.
 - Search over all parameters θ to minimize objective function using non-linear optimization.

Estimation algorithm: step 3

- Denote the mean utilities from step 2: $\delta_{jt}(\Gamma, \Sigma)$
- Compute $\tilde{\zeta}_{jt}(\theta) = \delta_{jt}(\Gamma, \Sigma) - x_{jt}\beta_0 - \alpha_0 p_{jt}$
 - Above equation is why we called Γ, Σ 'nonlinear' variables, and β_0, α_0 the 'linear variables'
- Interact with the instrumental variables to get the GMM objective function (denoting W as the GMM weight matrix):

$$\tilde{\zeta}(\theta)'ZWZ'\tilde{\zeta}(\theta)$$

- Solve for the parameters using nonlinear optimization.

$$\hat{\theta} = \arg \min_{\theta} \tilde{\zeta}(\theta)'ZWZ'\tilde{\zeta}(\theta)$$

- Note: since this is just a GMM problem, can also get standard errors using standard GMM methods

Numerical issues (documented by Knittel and Metaxoglou (2014))

- (See Conlon and Gortmaker (2020) for latest updates on best practices.)
- 1. Objective function is highly nonlinear with many local minima
 - Numerical results can be sensitive to starting values or choice of optimizer method
 - (Partial) solution: test results with different starting values and optimizer methods
 - (Partial) solution: choose an optimizer that is used for commercial purposes (e.g. Knitro)
 - (Partial) solution: Conlon and Gortmaker (2020) make some suggestions of free optimizer methods that work well in Scipy (a Python library)
 - Solution (probably not yet computationally feasible): use a global optimizer like 'differential evolution'
- 2. Need to choose very tight convergence tolerances for the inversion ($< 10^{-12}$)
- **These are common issues in structural models, so be on the lookout in other contexts.**

Alternative algorithm: MPEC ('Mathematical Programming With Equilibrium Constraints')

$$\begin{array}{ll} \min_{\theta, \zeta} & \zeta' Z W Z' \zeta \\ \text{subject to} & \tilde{\sigma}(\delta(\zeta); x, p, \hat{F}, \theta) = s \end{array}$$

- Notice minimization is over both θ and ζ here.
- Advantage of this approach:

Alternative algorithm: MPEC ('Mathematical Programming With Equilibrium Constraints')

$$\begin{array}{ll} \min_{\theta, \zeta} & \zeta' Z W Z' \zeta \\ \text{subject to} & \tilde{\sigma}(\delta(\zeta); x, p, \hat{F}, \theta) = s \end{array}$$

- Notice minimization is over both θ and ζ here.
- Advantage of this approach: No need for inversion step
- Disadvantage of this approach:

Alternative algorithm: MPEC ('Mathematical Programming With Equilibrium Constraints')

$$\begin{array}{ll} \min_{\theta, \zeta} & \zeta' Z W Z' \zeta \\ \text{subject to} & \tilde{\sigma}(\delta(\zeta); x, p, \hat{F}, \theta) = s \end{array}$$

- Notice minimization is over both θ and ζ here.
- Advantage of this approach: No need for inversion step
- Disadvantage of this approach: Many more parameters to solve for (ζ)
- Dube et al (2012): claim this approach results in a speedup.
 - However, can be complicated to program, and some have found it slow for large problems

Plan

1. Estimation algorithm
2. **Instrumental variables**
3. Extensions
4. Applications

Instruments

- We used the following moment conditions restriction:

$$E(\xi_{jt}|\mathbf{Z}_t) = 0$$

- Here, $\mathbf{Z}_t$ is a vector of instruments
- Note that above assumption implies a large set of potential IVs $z_{jt} = A_j(\mathbf{Z}_t)$ for which the unconditional moment restriction holds:

$$E(z_{jt}\xi_{jt}) = 0$$

- What instruments should we use for $\mathbf{Z}_t$?
 - Recall the dual role for instruments in the model.

Instruments: BLP instruments

- Use **characteristics of products in the market**.

$$E(\xi_{jt} | \mathbf{x}_t) = 0$$

- $\mathbf{x}_t$ is the vector of product characteristics in market t
- (Note: don't include price in characteristics)
- 'Observed characteristics mean independent of unobserved characteristics'
- Can form many moment conditions from above assumption. BLP use:
 - characteristics of own product
 - sum of characteristics of other products produced by the firm
 - sum of characteristics of competitors

Instruments: BLP instruments

- Power: closeness in characteristic space affects markups which affects price.
- Justification for this instrument:

Instruments: BLP instruments

- Power: closeness in characteristic space affects markups which affects price.
- Justification for this instrument: **product characteristics are set before ξ_{jt} known.**
 - Do you think this is a reasonable assumption in the car market?

Instruments: BLP instruments

- Power: closeness in characteristic space affects markups which affects price.
- Justification for this instrument: **product characteristics are set before ξ_{jt} known.**
 - Do you think this is a reasonable assumption in the car market?
- What if firms are forward looking and anticipate ξ_{jt} when choosing product characteristics?
 - Possible solution: use panel data.
 - E.g. Sweeting (2013) assumes $\xi_{jt} = \rho \xi_{jt-1} + u_{jt}$, where u_{jt} unanticipated at time $t - 1$
 - Implies moments: $E(\xi_{jt} - \rho \xi_{jt-1} | x_{t-1}) = 0$.
 - Comment: many connections here to the dynamic panel / production function literatures.

Instruments: cost-based / Hausman instruments

- Ideal instrument: cost-shifters
- But, cost data are rarely observed in practice.
- Hausman (1996) and Nevo (2001) use indirect cost measures: **prices in other markets**
 - i.e. $p_{jt'}$ for $t' \neq t$
 - Validity condition: conditional on x_t and $x_{t'}$, pricing is independent across markets and ζ_{jt} and $\zeta_{jt'}$ are independent.
 - In words: “IVs exploit common cost shocks across markets”
- Problems (example):
 - Unobserved advertising campaigns

Instruments: Waldfoegel-Fan instruments

- Used in Waldfoegel (2003), Fan (2013)
- Use **demographics in other counties where the product is sold**
 - Fan (2013): newspapers sold in multiple counties, uses demographics in other counties as IVs.
 - Idea: rely on consumption/preference externalities
 - E.g. Product offered in multiple counties $\rightarrow$ characteristics of product impacted by the attributes (like demographics) of the other counties.
 - Validity: conditional on variables in model, ξ_{jt} not correlated across counties (same assumption in Hausman instruments)
 - Additional concern: set of counties where product is offered is not exogenous

Plan

1. Estimation algorithm
2. Instrumental variables
3. **Extensions**
4. Applications

Extensions/other useful data sources

- Second choice data
- (e.g. from a survey)
- e.g. see Berry, Levinsohn and Pakes: “Differentiated products demand systems from a combination of micro and macro data: the new vehicle market” (2004)
- Micro-moments

Plan

1. Estimation algorithm
2. Instrumental variables
3. Extensions
4. **Applications**

Application: distinguishing between models of competition

- Nevo “Measuring Market Power in the Ready-to-eat Cereal Industry” (Econometrica, 2001)
- RTE Cereal
 - Concentrated market
 - High margins
 - High advertising-to-sales ratios
 - Aggressive introduction of new products
- “Used as a classic example of a concentrated differentiated-products industry in which price competition is approximately cooperative and rivalry is channeled into advertising and new product innovation”



Application: distinguishing between models of competition

- **Research questions**
- 1. Is there collusive pricing?
- 2. Decompose price-cost margins (PCM) into:
 - i. Product differentiation
 - ii. Portfolio effects (firms offer multiple products)
 - iii. Price collusion
- **Main takeaways for this class:**
 - See demand estimation 'in action'
 - Inclusion of a supply side that allows for horizontal competition
 - (Application/question is of general interest - but very "traditional IO")

Application: distinguishing between models of competition

- **Overall strategy:**
- 1. Estimate demand
- 2. Use estimates + pricing rules implied by different models of firm conduct to get price-cost margins (PCM)
 - Challenge: costs not observed
- 3. Compare predicted PCM from different models of conduct to true PCM
 - See which model of firm conduct best matches the data.
- **Main finding:**
- Nash-Bertrand pricing best matches observed PCM